Chapter 1

Introduction to the climate problem (short form)

Facts are stubborn things; and whatever may be our wishes, our inclinations, or the dictates of our passions, they cannot alter the state of facts and evidence.

—John Adams

Chapter overview

- Following Nicholas [4], we summarize the current state of knowledge about the climate problem in a few phrases:
  - It's warming.
  - It's us.
  - We're sure.
  - It's bad.
  - We can fix it (but we'd better hurry).
- These conclusions are based on some of the most well-established principles in physical science, corroborated by the work of thousands of scientists and many lines of independent evidence, measurements, and analysis. Every reputable academy of science, the United Nations, and the vast majority of the world's national governments and multinational corporations agree with these core climate truths.
- Keeping Earth's average temperature from rising much more than 1.5 °C from pre-industrial levels will require immediate, rapid, and sustained greenhouse gas emissions reductions. It will also almost certainly require the removal of carbon pollution from the atmosphere.
- Society should aim to hit 'net zero' greenhouse gas emissions globally by 2040 at the latest, sooner if possible, and every year thereafter should be 'net climate positive' until enough climate pollution is removed to reverse climate damages.
- Governments, companies, investors, and communities that can move faster should do so, and their early action will drive deployment-related cost reductions associated with learning-by-doing and economies of scale that will benefit the entire world.

doi:10.1088/978-0-7503-4032-8ch1

1.1 Introduction

Climate change is a complex issue, but the basic outlines of the problem are well known. In her book *Under the Sky We Make* sustainability scientist and author Kimberly Nicholas summarizes the climate issue in a few simple phrases [4]:

- It's warming.
- It's us.
- We're sure.
- It's bad.
- We can fix it.

Michael E Webber, a professor of energy resources at the University of Texas at Austin, rightly adds: 'But we'd better hurry'[1].

This chapter explains in a compact way why aggressive climate action is urgent, using Nicholas's framing (with Webber's addition) to structure the discussion of climate solutions. We provide a more elaborate version of these arguments in appendix A, with lots more graphs, which you can use as a resource when challenged on some of these key points by purveyors of misinformation. However, given the overwhelming scientific evidence pointing to the causes and magnitude of our climate crisis, much of the honest debate has shifted to *how* we should solve our climate challenges and *who* should be responsible for doing the solving. With that said, it's important to understand the basics of these core climate truths, as these facts underpin the urgency of action.

1.2 It's warming

Scientists have used many methods to estimate global average temperatures. Reliable and comprehensive direct measurements began around 1850, but proxy methods, using tree rings, ratios of isotopes, and other techniques, can yield estimates going back thousands or even millions of years. Of course, the further back we look, the more complicated it is to do accurate assessments, but even the proxy estimates are based on well-established principles of physical science.

Figure 1.1 shows a summary of the instrumental temperature record since 1850, created by researchers at the University of Oxford [5] and continually updated. These data combine four well accepted time-series of temperatures into what they call a **climate change index**. Such temperature data are expressed relative to a base year (or an average over a set of years).

This graph is expressed as a change in temperature relative to a 1850–1900 baseline, which is one way to characterize what we call a **pre-industrial baseline**. The industrial revolution, which was the beginning of large-scale exploitation of fossil fuels, started in the late 1700s, but didn't really gather steam until the late 1800s. That fact combined with widespread instrumental temperature data only

[1] https://twitter.com/MichaelEWebber/status/1335950979334893572?s=20

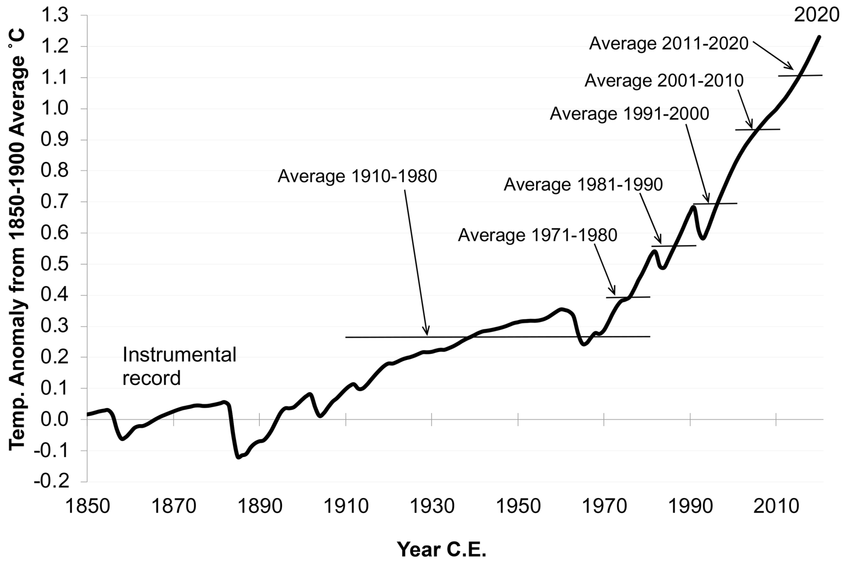

Figure 1.1. Global average temperatures, 1850–2020. Source: Based on data through 2020 from https://globalwarmingindex.org, first assessed and presented in [5].

extending back to about 1850 have led scientists to use the 1850–1900 baseline in the most recent research.

Temperature changes in the 1800s were mostly driven by volcanoes and other natural forces. The effect of Krakatoa in 1883 was particularly dramatic, with volcanic ash cooling the Earth by more than 0.1 °C over a year or two. Temperatures in the 1900s were also affected by such events but increased by more than 1 °C by 2020 due mainly to increasing concentrations of greenhouse gases during this period (see below). Each decade since the 1970s has been hotter than the last, and almost all of the hottest years ever recorded have occurred in the past two decades[2].

Warming since pre-industrial times has pushed the Earth out of the comfortable and stable temperature range in which human civilization developed [6, 7]. The *rate* of temperature change in the last century (as well as the rate of change in the underlying drivers of temperature change) is also much more rapid than humanity and the Earth have experienced in thousands of years, which is another reason for concern, as we discuss below.

This warming is reflected in other indicators. Northern Hemisphere summer sea-ice extent has reached historically low levels in recent years, while Antarctic sea ice saw significant declines in the early to mid-twentieth century [8] with Antarctic sea-ice extent falling below long-term averages from 2015 onwards[3]. Sea levels keep increasing as land-based glaciers melt [9] and a warmer ocean expands [10, 11].

[2] https://climate.copernicus.eu/copernicus-globally-seven-hottest-years-record-were-last-seven

Sea level rise has also been accelerating in recent years [12, 13], at the same time as global average upper ocean heat content has increased rapidly [14]. Finally, satellite measurements of Earth's energy balance have confirmed that greenhouse gases are trapping heat, just as scientists expected [15].

It's important to understand that **global warming** does *not* mean that everywhere is breaking temperature records all the time. Natural cycles and fluctuations still exist, and places can still experience record cold temperatures on a given day. Many places are experiencing extremes in both directions from climate change, particularly the **weakening of the jet stream** in the Northern Hemisphere linked to melting Arctic sea ice [16], which can allow hot air to move further north and cold air to move farther south (contributing to localized **polar vortex** cold snaps). However, despite localized variability, average annual and daily high temperatures are increasing almost everywhere we look.

We have even longer-term data for historical temperatures going back millions of years, as shown in Burke *et al* [17]. The uncertainties grow as we move back in time, but it is clear Earth was a lot cooler (−3 °C to −5 °C from our 1850 to 1900 baseline) for most of the 300 000 years before the present, with kilometers-deep ice sheets covering much of Earth's land masses.

Temperatures about 3 million years ago were about at about current levels, and before that (50–60 million years ago) temperatures were much hotter (10 °C to 15 °C above 1850–1900). Sea levels in that warmer period were more than one hundred meters higher than they are today [18].

Multiple independent lines of evidence are consistent with a rapidly warming Earth. The world's foremost climate science authority, the **United Nations Intergovernmental Panel on Climate Change** (IPCC) concluded in 2021 [19] that 'global surface temperature has increased faster since 1970 than in any other 50 year period over at least the last 2000 years'.

1.3 It's us

We know why average temperature have increased over the past century: emissions of **greenhouse gases** (GHGs) related to human activity have increased substantially since pre-industrial times, leading to increasing **concentrations** of these gases in the atmosphere. GHGs such as **carbon dioxide** (CO_2) trap energy from the Sun that otherwise would radiate to space, acting like a blanket that has been getting thicker for more than two centuries. To first approximation, warming to date is 100% driven by human activities.

Global temperature changes have historically corresponded very closely with atmospheric GHG concentrations [19]. The basic science behind how greenhouse gases warm the Earth through the **greenhouse effect** has been understood since the 1800s, and pre-industrial levels of atmospheric GHGs have been instrumental in keeping Earth warm enough for life to evolve. But while GHG levels have naturally

[3] https://nsidc.org/data/seaice_index/

fluctuated over millennia, the rapid rise in greenhouse gases since 1900, driven by human activities, is unprecedented in Earth's history.

Figure 1.2 shows man-made emissions of all greenhouse gases from 1850 to 2020, which have increased rapidly and exponentially. There are four major categories of greenhouse gases: carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), and other gases (mostly what are called 'F-gases' that contain fluorine). The non-CO_2 gases are converted to what's called **CO_2 equivalent** (CO_2e), which is the equivalent amount of CO_2 that would result in the same warming effect as emissions of these other gases over a 100 year period. This technique allows us to approximate the total warming effect for all gases over time.

Total greenhouse gas equivalent emissions have increased at a rate of 2.2%/year from 1850 to 2020, with the emissions from fossil energy use growing at a 3.1%/year

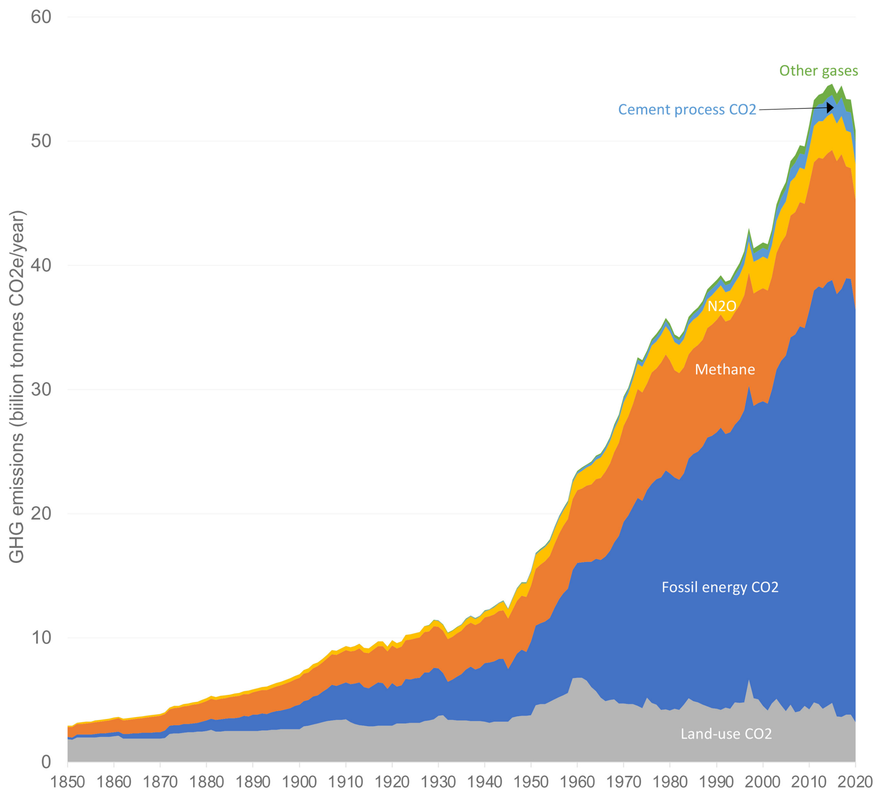

Figure 1.2. Greenhouse gas emissions expressed as CO_2 equivalent 1850 to 2020. Source: Fossil, cement, and net land-use CO_2 emissions 1850 to 1958 from CDIAC archives: https://cdiac.ess-dive.lbl.gov/trends/emis/tre_glob_2013.html. Fossil, cement, and net land-use CO_2 emissions 1959 to 2020 from [20]. Fossil CO_2 emissions include combustion from flaring, solids, liquids, and gases. Methane, N_2O, and F-gas emissions from 1850 to 2019 from PIK, taken from https://www.climatewatchdata.org/data-explorer/historical-emissions. Global warming potential values for methane and N_2O adjusted to reflect AR6 100 year values in table 2.1 (PIK data use AR4 values as described in [21]). Emissions of methane, N_2O, and F-gases for 2020 estimated assuming emissions stay at 2019 levels (just like they did during the 2009 recession).

rate over that period. There have been periods of modest declines, but the overall upward trend has been inexorable.

The most important warming gas is carbon dioxide, which comes mainly from combustion of fossil fuels for energy, but also from deforestation and production processes for cement, steel, aluminum, and other materials. The burning of fossil fuels and land-use changes are the main sources adding CO_2 to the biosphere. The ocean and land have historically taken up about half of these emissions, but what remains goes into the atmosphere and stays there for centuries. As these emissions continue over time, the concentrations of CO_2 in the atmosphere go up, **climate feedback loops** (such as wildfires and melting permafrost) can further accelerate GHG emissions and associated warming.

Scientists have different methods to assess past concentrations of trace gases over time. One way is to drill for ice cores and extract and analyze air samples from bubbles in the ice. For more recent times (since 1959) we have detailed direct measurements of atmospheric concentrations from the observatory on Mauna Loa in Hawaii and other locations.

Over the last 800 000 years, Earth's atmosphere never held more than about 300 **parts per million** of CO_2 [22] corresponding to significantly cooler temperatures than in the Holocene (the twelve thousand year period before recent rapid human-caused warming) as shown in Burke *et al* [17]. In 2020 we hit 414 parts per million, and in May of 2022 we hit 421 parts per million. The increase over historical levels has occurred mostly in the span of about one and a half centuries. When it comes to CO_2 concentrations, we've moved rapidly into uncharted territory [23], driven by the increasing CO_2 emissions shown in figure 1.2.

Total CO_2 equivalent concentrations (including all warming agents) reached just over 500 parts per million (ppm) in 2020[4]. For comparison, concentrations in 1850 were about 305 ppm. That means current concentrations are now about 1.6 times pre-industrial levels.

The IPCC in 2021 tallied the main drivers of warming to the 2010 to 2019 period, relative to the 1850 to 1900 period [19]. Those results are shown in figure 1.3. Higher positive numbers mean more warming effect, and negative numbers mean that factor causes cooling, mainly by reflecting sunlight back to space rather than trapping it as heat.

The most important driver of increases in global surface temperatures since pre-industrial times has been increasing concentrations of carbon dioxide, mostly from combustion of fossil fuels, with a significant additional contribution from changes in land-use patterns (such as deforestation). Next is methane, driven by land-use changes, energy production, and agriculture, followed by non-methane volatile organic compounds and carbon monoxide, halogenated gases (such as the chlorofluorocarbons being phased out under the Montreal Protocol for ozone depleting chemicals, plus some others), nitrous oxides (also mainly from agriculture), black carbon, and airplane contrails. The cooling agents are sulfur dioxide

[4] https://gml.noaa.gov/aggi/aggi.html

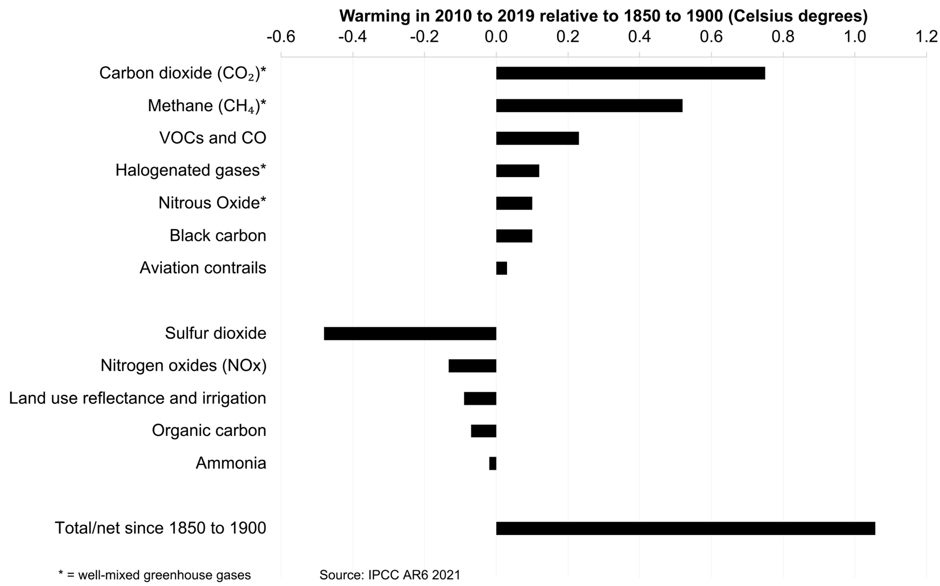

Figure 1.3. Contributors to warming in 2010 to 2019 compared to the 1850 to 1900 baseline. Source: ICPP Working Group I, Summary for Policy Makers, Sixth Assessment Report [24].

aerosols (small particles that reflect sunlight), nitrogen oxides, land-use reflectance and irrigation, organic carbon, and ammonia.

The cooling effects just about cancel out the effect of other factors than CO_2 and half of the warming associated with methane, but that doesn't mean that we can ignore the other warming agents. As we start to phase out fossil fuels, the cooling effects from the aerosols will also be reduced, so we'll need to compensate by also rapidly minimizing the shorter-lived warming agents such as methane, some fluorinated gases, and black carbon.

1.4 We're sure

Uncertainties always exist in science, but when we confirm scientific findings by multiple, independent lines of evidence, we call them 'facts'. There may still be uncertainty about some details, but the preponderance of the evidence points towards these facts accurately describing how the physical world operates. Every credible scientific organization now agrees that burning fossil fuels and other human activities are almost entirely responsible for current climate change, based on decades of research from thousands of scientists. Human-caused climate change is now as much of a settled scientific fact as gravity.

The US National Academy of Sciences [25], which is not known for its wild speculation, concluded in 2010:

> A strong, credible body of scientific evidence shows that climate change is occurring, is caused largely by human activities, and poses significant risks for a broad range of human and natural systems….
>
> Some scientific conclusions or theories have been so thoroughly examined and tested, and supported by so many independent observations and results, that their likelihood of subsequently being found to be wrong is vanishingly small. Such conclusions and theories are then regarded as settled facts. This is the case for the conclusions that the Earth system is warming and that much of this warming is very likely due to human activities.

The academies of science for 80 other countries (including China, India, Russia, Germany, Japan, Brazil, and the UK) have released comparable statements about the science of climate [26]. The IPCC, the global scientific body charged with investigating this issue, stated with uncharacteristic bluntness in 2007, that 'warming of the climate system is unequivocal, as is now evident from observations of increases in global average air and ocean temperatures, widespread melting of snow and ice and rising global average sea level' [27]. IPCC reports in 2013 [28] and 2021 [19] were even more emphatic, as was the World Meteorological Association in 2021 [29].

That greenhouse gases warm the Earth is a finding based on some of the most well-established principles in physical science, as well as extensive measurements. The largest historical source of warming is carbon dioxide, and we know for a fact that the increase in carbon dioxide concentrations after the industrial revolution (particularly after the mid-1800s) was fueled predominantly by human activity.

One reason that we know that fossil fuels and land-use changes are the cause of the measured increases in CO_2 concentrations is because the total carbon emitted since the dawn of the industrial age is about twice as large as the total amount that remains in the atmosphere nowadays (the other half was absorbed by the oceans and land-based biota), and there are no other known sources of carbon that could account for such an increase in the atmosphere's CO_2 content.

In addition, scientists can measure the prevalence of different isotopes of carbon in the atmosphere to understand the sources of CO_2. Carbon from recent biological activity (such as deforestation) contains a different mix of carbon isotopes than carbon from fossil fuels and other geological sources, and scientists have measured changes in the concentrations of those isotopes in the atmosphere that are consistent with the additional carbon coming from human-linked sources [30]. So it's virtually certain that humans are the cause of elevated CO_2, as well as causing similar increases in the past two centuries in concentrations of methane, nitrous oxides, and other GHGs [19].

These changes are warming the planet, as shown in figures 1.1 and 1.3, and as predicted with surprising accuracy as early as 1975 in *Science* [31] and in 1982 by scientists at Exxon [32]. The best current estimates are that global average surface temperatures have increased 1.1 °C–1.2 °C since 1900.

The physical science results supporting these conclusions go back at least a century to the Swedish scientist Svante Arrhenius, who calculated in 1896 the first climate sensitivity of 4 °C to 5 °C for a doubling of carbon dioxide concentrations. Arrhenius's analysis was supported by earlier measurements made by Eunice Foote and John Tyndall that demonstrated the heat trapping abilities of CO_2 [33]. The idea that greenhouse gases could warm the Earth is not a new one, and in fact the first informed speculation about this topic was by the mathematician Joseph Fourier in the 1820s [34].

These concerns are also validated by actual measurements of the climate system that corroborate theoretical predictions [35]. Satellite measurements show, for example, that as greenhouse gases have built up in the atmosphere, the heat emitted from the Earth has been declining at wavelengths that exactly correspond to those absorbed by various GHGs [36]. Similar measurements show that thermal radiation back to Earth's surface from the atmosphere, which we'd expect to increase if greenhouse gases trap heat, has been increasing exactly as we thought it would [37]. And the amount of heat stored in the oceans over the past few decades has been rising rapidly, which is consistent with a warming planet [24, 38, 39][5].

We can also examine data on some key indicators of warming, which by most accounts are changing at rates equaling or exceeding our worst-case predictions of just a few years ago [19, 40, 41]. These measurements are one of the main reasons why scientists are so alarmed about humanity's effect on the planet's temperature.

In summarizing these and other measurements, the IPCC [27] concluded in 2007 that 'observational evidence from all continents and most oceans shows that many natural systems are being affected by regional climate changes, particularly temperature increases'. By 2021, the IPCC [24] was even more explicit: 'Human-induced climate change is already affecting many weather and climate extremes in every region across the globe. Evidence of observed changes in extremes such as heat waves, heavy precipitation, droughts, and tropical cyclones, and, in particular, their attribution to human influence, has strengthened since AR5 [in 2013]'.

The observations cited above are powerful evidence for a warming world, but they are not the only ones [19]. Glaciers are melting [9, 11], wildfires are more frequent and more destructive [42, 43], agricultural growing seasons are changing [44], agricultural productivity is down [45], wildlife are altering their long-established patterns [46], and billions of people are (or will soon be) experiencing new heat extremes [47, 48]. Something's clearly happening to the climate, and these indicators are consistent with what the last one and a half centuries of science has been saying about this problem all along.

1.5 It's bad

If we continue on our current trajectory, we'll push Earth's climate into uncharted territory [49]. Figure 1.4 shows the result from one widely cited 'current trends continued' case (also known as SSP-2) created around 2015 or so [50]. For comparison, we also plot the instrumental record from figure 1.1 and the current best estimates for temperatures over the past twelve thousand years, from [6, 7].

[5] Also see https://www.climate.gov/news-features/understanding-climate/climate-change-ocean-heat-content.

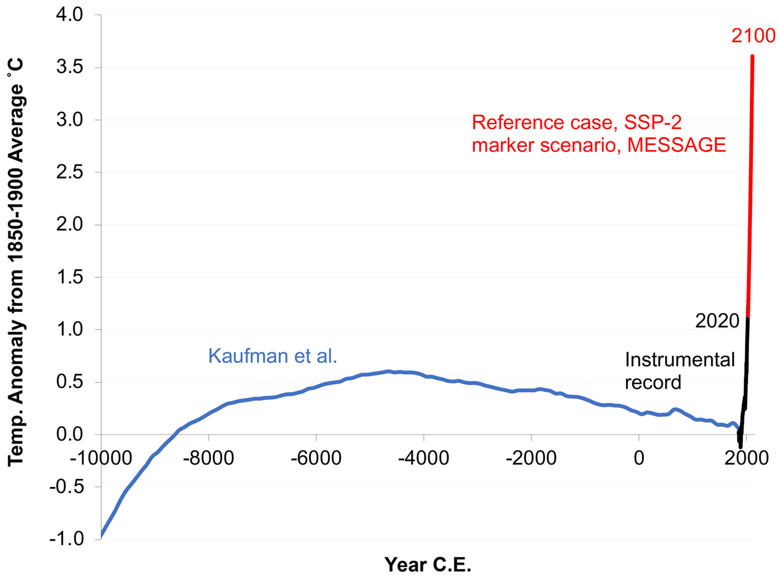

Figure 1.4. Historical temperatures contrasted with the reference-case projection from a prominent 2017 study. Source: Kaufman *et al* [6] for the past 12 000 years [6], https://globalwarmingindex.org for the instrumental record [5], and Fricko *et al* [50] for the reference-case, SSP-2 marker scenario [59].

That emissions trajectory for greenhouse gas emissions (and thus, global temperatures) results in an increase of about 3.6 °C above pre-industrial levels by 2100. This result is typical for assessments of a 'current trends continued' path circa 2015, which fall in the range of 3 °C to 4 °C above pre-industrial times. In the past few years, with significant climate action in some major countries, the 'current trends continued' case has improved even more [51–54] to more like 2.5 °C to 3.5 °C.[6]

In 2012 when one of us (Koomey) wrote a book assessing the options for reducing emissions [55], the current path at that time implied an increase of about 5 °C by 2100. Since 2012 we've made great progress in bringing down the cost of clean technology, we've shut down many coal plants, we've implemented and strengthened policies to reduce emissions, and we've realized that some of the more dire projections circa 2010 were based on overestimates of exploitable reserves of coal [56]. Of course, there is great uncertainty in any projection into the future, but we need a benchmark against which to measure progress, and 2.5 °C to 3.5 °C by 2100 is as good an estimate as any.

That improvement is small comfort, however. Even our current trajectory would be a disaster for the Earth and most other species [49, 57]. It would raise Earth's average temperature to a level not seen for about four million years [17], and to do so with unprecedented speed.

The most important direct effect of increasing temperatures is stress on natural and human systems [49]. Greenhouse gases keep more energy in the climate system,

[6] Also see https://thebreakthrough.org/issues/energy/3c-world

and that energy must go somewhere. Where it goes is into extreme rainfall and temperature events, which will become increasingly difficult or impossible for humans to manage, and ever more devastating for natural systems (whose ability to adapt is even more limited) [58].

Figure 1.5 illustrates how a warming climate 'loads the dice' and makes extreme temperature events more likely and the extremes more extreme, just as more moisture in the air makes high precipitation events more likely (and oddly enough, makes droughts in some places more likely and more intense as well) [24, 48, 60–64]. It also makes wildfire much more likely [43].

This concern is not a theoretical one. In a NASA report published in 2012, James Hansen and his colleagues examined distributions of measured temperatures for the period 1951 to 1980, comparing them to those from the 1980s, the 1990s, and the 2000s [65]. In each succeeding decade, the shifting of the distribution to towards higher temperatures became more pronounced as the climate warmed.

As we showed in figure 1.4, the current global environment is accustomed to a relatively narrow temperature range, one that has prevailed for thousands of years. Ecosystems can sometimes migrate slowly (over many millennia, not decades or centuries), but that migration is limited by geography and other constraints. For example, a forest biome can gradually move up the mountainside as the climate warms (soil and geography permitting), but once it reaches the peak there's nowhere else to go, and extinction is the result. On our current path, thousands of plant and animal species that have existed for eons will be driven to extinction in the span of a century or so [66–68].

Humans and their support systems are also vulnerable to a warming climate. Heat waves will become ever more frequent [69], and people in locations without

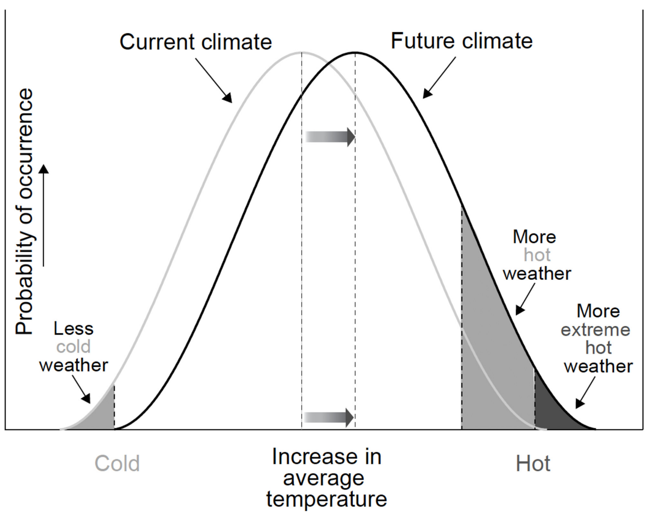

Figure 1.5. Increasing temperatures load the dice and make extreme temperatures more likely (and the extremes more extreme). Source: Adapted from a graph made by the University of Arizona, Southwest Climate Change Network, http://www.southwestclimatechange.org. Reproduced with permission from [55].

air conditioning will either add it (which will worsen climate change), suffer, or even die (as tens of thousands of Europeans did during the heat wave of 2003). Our wastewater treatment and water supply systems are designed to handle current conditions but will be difficult and expensive to adapt to a warming world's rapidly rising sea level and increasingly intense rainstorms. Wildfires will become more frequent and more intense [43]. Pollen and its associated respiratory effects will worsen [70]. Climate change will also increase risks of cross-species viral transmission [71, 72].

With even small increases in sea level, low lying coastal areas will become increasingly vulnerable to storm surges, putting millions of lives at risk, particularly in the developing world. Those areas will also suffer from increased saltwater intrusion into groundwater supplies. A 'current trends continued' path implies more than 0.5 m rise in sea level by 2100 [24, 73–75], which would represent significant challenges to human society, and sea level rise will continue for centuries, barring substantial carbon removal from the atmosphere.

There are also indirect effects. One of the most important is an increase in the acidity of the oceans, caused by more dissolved CO_2 (which creates carbonic acid). This development will make life increasingly difficult for many types of aquatic life, with rates of acidification proceeding more rapidly than at any time in the past 65 million years [76]. The acidification effect (plus the increase in ocean temperatures) means that coral reefs will likely be a thing of the past by the end of the twenty-first century, and will pose increasing challenges to marine life of all types [77, 78]. It also is one reason why schemes such as those proposed to inject particles into the atmosphere to cool the Earth are ultimately chimerical—as long as more CO_2 dissolves into the oceans, the acidification effect will intensify, and just reflecting more sunlight won't fix it.

Remember also that the climate sensitivity measures the *average* temperature change for a doubling of greenhouse gas concentrations. Changes at the Earth's poles have been [79] and will be much larger (that's just how the system works). The most likely case for climate sensitivity combined with the reference-case emissions forecast would ultimately lead to an ice-free planet Earth and sea level rises much bigger than even recent projections. It also means that large releases of carbon trapped in the permafrost and in methane hydrates beneath the ocean floor are much more likely, and that would amplify the warming effect.

Perhaps the most worrying aspect of climate change is the unknowable but non-zero probability that pushing the Earth's climate out of its recent equilibrium might lead to 'tipping points', discontinuous change, and catastrophic disruptions of weather and climate [80–82]. One example that has preoccupied scientists for decades is the possibility that melting ice could disrupt the Gulf Stream in the North Atlantic [83], and there has been evidence of a weakening of those currents in recent decades [84–86]. The rapid release of carbon and methane from melting permafrost and methane from warming oceans are two more. The probability and consequences from catastrophic events are inherently *unknowable*, and that reality disrupts the standard benefit–cost model for assessing the economics of climate mitigation [82, 87–90].

1.6 We can fix it (but we'd better hurry)

Let's start by defining what we mean by 'fixing' climate change. Solving climate change starts with stabilizing global surface temperatures at the lowest possible level above pre-industrial times, and that means *reducing emissions to zero as soon as possible* [91].

Many climate action plans now focus on getting climate pollution to **net zero**, the point where GHG emissions are reduced far enough that any remaining remissions are counterbalanced by GHG emission removals from natural and engineered systems [92]. Net zero by 2050 (or sooner) targets have now been adopted by many governments, as well as many of the world's largest companies and investors, catalyzed by goals of the United Nation's 2015 **Paris Agreement**.

Unfortunately, net zero is not enough. Even if Earth gets to net zero GHG emissions tomorrow, without additional action we're still locked into at decades of additional warming, and a new climate equilibrium that leaves the world worse off than it is today. Droughts, floods, heat waves, wildfires, rising seas, crop failures, and other climate-linked disasters will continue to get worse. Many more lives and livelihoods will be lost, and some once thriving places may become unlivable.

To truly solve climate change, we must view net zero as a transition point rather than an end goal. We should aim instead to get our planet to net **climate positive**, sometimes also called **carbon negative**, where there is a net removal of carbon from the atmosphere every year. Our collective goal should be to make Earth climate positive until we have returned the climate to a state that's as close as possible to the pre-industrial range for which human civilization and the ecosystems around us have evolved. We should also aim to repair the damages caused by climate change as much as we can and compensate those who have suffered when damages can't be repaired. Fixing climate change requires our best efforts to not just stop the bleeding, but also heal the wounds.

This doesn't mean that we shouldn't aim to reach (and surpass) net zero as soon as possible, but net zero by 2050 is not soon enough, as we discuss below. Because of the long residence time of the most important greenhouse gases in the atmosphere, warming is to first approximation proportional to *cumulative* emissions [93, 94]. That means every molecule emitted matters, and that if we stop emitting greenhouse gases, warming will eventually stop [95, 96]. This makes the climate problem different from other types of air pollution, which generally stay in the atmosphere for a much shorter time than do emissions of carbon dioxide, nitrous oxide, and many of the other gases.

The importance of cumulative emissions means we have no time to lose in reducing our emissions, a fact that is not widely enough appreciated. We've already dithered for more than three decades as emissions kept increasing, and every day we delay getting to climate positive makes the situation worse [91].

The good news is that we already have almost all the climate technologies we need, and these solution technologies are improving every day. The same technologies that can get us to net zero can propel us beyond to climate positive and a regenerative climate future. The biggest climate challenges are sociological, political, and economic rather than technological. Society has plenty of money to pay for the transition but shifting this money out of climate pollution and into solutions requires navigating powerful entrenched interests, perverse incentives, bureaucratic barriers, and outdated ways of thinking. Solving climate change will also require much better communications about our climate impacts and solutions, engaging all levels of society in shifting to a climate-positive economy. We can do it, but we need to decide to do it. We'll get into specific recommendations in the following chapters.

1.6.1 What is a warming limit?

The warming limit approach has its origins in the realization that stabilizing the climate at a certain temperature to minimize climate risks (e.g. a warming limit of 1.5 °C or 2 °C above pre-industrial times) implies a particular emissions budget, which represents the total cumulative greenhouse gas emissions compatible with that temperature goal [97]. That budget also implies a set of emissions pathways that are well defined and tightly constrained (particularly now that we've squandered the past three decades by not reducing emissions). This risk-minimization approach, which can also be described as 'working toward a goal', also involves assessing the cost effectiveness of different paths for meeting the normatively determined target [55].

A warming limit is more than just a number (or a goal to be agreed on in international negotiations). It embodies a way of thinking about the climate problem that yields real insights [98]. A warming limit is also a value choice that is informed by science. It should not be presented as solely a scientific 'finding', but as a value judgment that reflects our assessment of societal risks and our preferences for addressing them.

The warming limit approach was first suggested for externalities more generally in [99] and was explored (then quickly dismissed) by the climate economist William Nordhaus [100, 101]. It had its first fully developed incarnation in 1989 in [1] (which was subsequently republished in 1992 [2]). It was developed further in [102] and [103], and served as the basis for the International Energy Agency's analysis of climate options in 2010, 2011, 2012, and 2020 [104–107].

This way of thinking was developed as a counterpoint to the prevailing 'benefit–cost' approach favored by the economics community, and it has many advantages [98]. It encapsulates our knowledge from the latest climate models on how cumulative emissions affect global temperatures, placing the focus squarely on how to stabilize those temperatures. It places the most important value judgment up-front, embodied in the normatively determined warming limit, instead of burying key value judgments in economic model parameters or in ostensibly scientifically chosen concepts such as the discount rate. It gives clear guidance for the rate of emissions reductions required to meet the chosen warming limit, thus allowing us to determine if we're 'on track' for meeting the temperature goal and allowing us to adjust course if we're not hitting those near-term guideposts.

The warming limit approach also allows us to estimate the costs of delaying action or excluding certain mitigation options and provides an analytical basis for discussions about equitably allocating the emissions budget. Finally, instead of pretending that we can calculate an 'optimal' technology path based on guesses at mitigation and damage cost curves decades hence, it relegates economic analysis to the important but less grandiose role of comparing the cost effectiveness of currently available options for meeting near-term emissions goals [98].

The warming limit approach shows that delaying action is costly, required emissions reductions are rapid, and most proved reserves of fossil fuels will need to stay in the ground, and large amounts of carbon will need to be removed from the atmosphere if we're to stabilize the climate and repair the damage. These ideas are familiar to some, but many still don't realize that they follow directly from the warming limit framing:

- Delaying emissions reductions forecloses options and makes achieving climate stabilization much more difficult [108]. 'Wait and see' for the climate problem is foolish and irresponsible, which is obvious when considering cumulative emissions under a warming limit. The more fossil infrastructure we build now, the faster we'll have to reduce emissions later. If energy technologies changed as fast as computers there could be justification for 'wait and see' in some circumstances, but they don't, so it's a moot point.
- Absolute global emissions will need to turn down immediately and approach climate-positive territory in the next few decades [109] if we're to have a good chance to keep global temperatures 'well below 2 °C', as many in the scientific community advocate [110]. The emissions pathways given the current carbon budgets are tightly constrained. Even if the climate sensitivity is at the lowest end of the range included in IPCC reports (1.5 °C), that only buys us another decade in the time of emissions peak [111], which indicates that the findings on emissions pathways are robust, even in the face of uncertainties in climate sensitivity.
- The rate of emissions reductions, which is a number that can be measured, is one way to assess whether the world is on track to meet the requirements of a particular warming limit. We know what we need to be doing to succeed, and if we don't meet the tight time constraints imposed by that cumulative emissions budget in one year, we need to do more the next year, and the next, and the next. It's a way of holding policy makers' proverbial feet to the fire.
- The concept of 'stranded fossil fuel assets' that can't be burned, popularized by Bill McKibben [112] and Al Gore [113], follows directly from the warming limit framing. In fact, Krause *et al*'s 1989 book, *Energy Policy in the Greenhouse* [1] had a chapter titled 'How much fossil fuel can still be burned?', so the idea of stranded assets is not a new insight (but it is a profound one).

1.6.2 An evolution in thinking

The warming limit approach continues to be helpful in building the case for urgent action on climate [114–116] but its original policy usefulness rested on the overarching assumption that there was still time left to address the crisis. As time has passed and climate damages continue to mount it has become clear to us that the world is running out of time [117], and given uncertainties in estimating the carbon budget [118, 119], it is in our judgment better to focus on concrete emissions reductions goals instead of worrying too much about how much carbon budget is left.

The IPCC completed its report on scenarios based on a 1.5 °C warming limit in 2018 [110]. That study found, after reviewing many studies that meet the 1.5 °C warming limit, that cutting global greenhouse gas emissions at least in half from 2020 to 2030 (with subsequent reductions to follow at a similar pace) is a critically important milestone for success. This rough rule of thumb, which was enshrined as a 'carbon law' by Johan Rockstrom and his colleagues in a seminal article in *Science* in 2017 [109], describes the minimum level of effort needed to hit a 1.5 °C warming limit. Rockstrom's 'law' implies halving of absolute emissions in each decade starting in 2020, reaching close to net zero emissions by 2050.

When talking about rapid emissions reductions it is customary to refer to rates of change as a percentage of base year emissions, instead of often-used exponential rates of decline [55]. This convention is used because rates of decline reach astronomical levels in percentage terms as emissions approach zero, and percentages of a base year maintain an intuitive physical meaning throughout the analysis period. With this convention, a 5%/year decline in emissions relative to 2020 would result in a halving of annual emissions by 2030 (5%/year × 10 years).

A strong case can be made for an even more aggressive goal than implied by Rockstrom *et al* achieving net zero global emissions by 2040. That goal implies a rate of emissions reduction of 5%/year relative to 2020 continuing to 2040. We explain the rationale for such an aggressive goal in the next chapter.

When evaluating long-term goals such as these, it's important not to place too much emphasis on precise numbers, particularly many decades hence. It is valuable, however, to look at broader lessons from such scenarios, and the most important lesson is the rapid rate of change embodied in all scenarios that put a 1.5 °C warming limit in reach. We'll need to build zero emissions energy and industrial process technologies at high rates and retire existing high-emissions capital on a rapid schedule.

Achieving such speedy emissions reductions will require unprecedented changes in how the global economy generates value. In coming decades, many processes in our economy will need to be re-evaluated and re-designed from scratch to minimize or eliminate emissions.

1.6.3 Can it be done?

The question of whether a modern society can achieve such rapid reductions is one we'll explore in the rest of this book, but it cannot be answered precisely by modeling or analysis. We'll only actually know how far we can reduce emissions once we start trying to do so in earnest, and we really haven't started yet.

Every tenth of a degree matters. If we overshoot 1.5 °C, so be it, but 1.6 °C is much better than 1.7 °C, which is much better than 1.8 °C, and by getting to a climate-positive state we can start to bring temperatures back down and reduce damages from temporarily overshooting climate targets. Even if we are daunted by the challenge of keeping warming below 1.5 °C, we need to try, and we need to act as quickly as we can.

For climate change, 'moving in the right direction' isn't enough, as Solomon Goldstein-Rose points out in his excellent book *The 100% Solution*:

> There's a lot of rhetoric about 'moving in the right direction' on climate change. But because of its difference from other issues (impacts being caused not by each year's emission but by cumulative emissions until we start removing them from the atmosphere), there's not really such a thing as 'moving in the right direction.' Climate change impacts get exponentially worse until we solve the problem 100%. That's why it is so much scarier and more urgent than other problems…
>
> The idea of 'doing what we can' is dangerous when it comes to climate change because it implicitly accepts that the maximum viable action is less than the minimum needed action.

'Can it be done?' is therefore the wrong question, and worrying about feasibility is the wrong framing. The right question is 'How can we change society to do what is necessary?'

Nobody knows what's likely or even possible until we start down the path of aggressively reducing emissions by deploying technology, capital, communications, and institutional innovations at the requisite scale. If we choose to do so, many things will become possible that wouldn't be possible if we didn't.

Feasibility also depends on context, and on what we are willing to pay and prioritize to minimize risks. What if we finally decide (as we should) that it's a real emergency (like World War II)? In that case we'd make every effort to fix the problem, and what would be possible then is far beyond what we could imagine today.

It is therefore a mistake for analysts to impose an informal feasibility judgment when considering a problem such as this one, and instead we should aim for what we think is the best outcome from a risk-minimization perspective, and if we don't quite get there, then we'll have to deal with the consequences. But if we aim too low, we might miss possibilities that we'd otherwise be able to capture.

History shows that under the right conditions, societies and industries can move quickly. In the beginning of World War II, the US retooled much of its heavy industry over the span of about 6 months [120], and some other nations engineered similarly rapid change. We now have some technology advantages over industrial firms of that era [121], especially information technology, which is our 'ace in the hole' [122].

We also know that existing technology offers opportunities to reduce emissions substantially right now, and maximizing immediate emissions reductions is where we should be focused, not on worrying about whether we'll be able to get to zero emissions by 2040. Do the obvious things: shut down fossil fuel power plants (starting with coal), mandate electrification where possible, install much more wind, solar, and energy storage, and deploy all other existing emissions reduction technology at scale everywhere we can. Our choices now create our options later because deployment drives costs down, creating new opportunities for reducing emissions elsewhere.

1.6.4 Keeping carbon in the ground

Another way to think about how fast emissions need to come down is by comparing projected fossil fuel consumption to the latest estimates of proved reserves of fossil fuels. In geology parlance, proved reserves are those stocks of fossil fuels that are known to exist with high confidence and that can be extracted using current technology at current prices. Resources are fossil deposits known with less confidence and/or are not extractable using current technology and current prices.

The first bar in figure 1.6 shows proven fossil reserves for 2018 from [123]. They total about 900 billion tonnes of carbon (not carbon dioxide). The second bar shows emissions from a widely cited reference case [50] from 2015, which shows about 1300 billion tonnes of carbon combusted from 2020 to 2100. That result implies that a small fraction of the remaining resources (which are more than tenfold bigger than reserves) would need to be converted to reserves through exploration or using new technology to meet that demand.

The more important bar for our narrative, however, is the one for the low energy demand case [59], which is a scenario that would keep global temperatures from increasing no more than 1.5 °C from pre-industrial times. In that scenario, the world can burn less than one tenth of the fossil carbon implied in the reference case, and only one eighth of the proved reserves. That means we'll either need to keep almost 90% of global fossil reserves in the ground unburned or identify some way to sequester the carbon from burning it, which will be a heavy lift at the required scale.

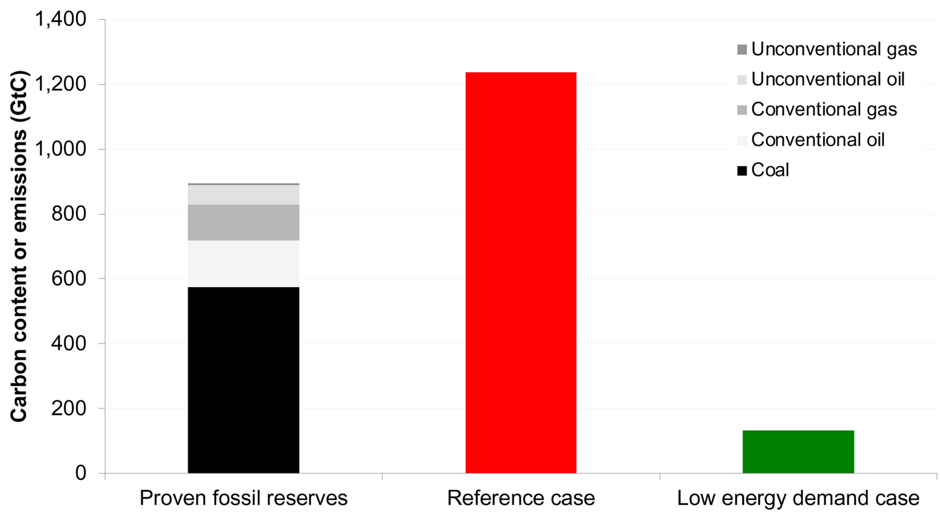

Figure 1.6. Comparing reference and low energy demand cases to proved fossil reserves. Source: BGR [123], Fricko *et al* [50], Grubler *et al* [59].

These exact numbers are dependent on some key assumptions in this scenario, but the main point is the same for all aggressive mitigation scenarios that keep the 1.5 °C warming limit in reach: *We'll need to keep a substantial fraction of proved fossil reserves in the ground or find another safe way to sequester that carbon.*

1.6.5 The economics of climate action

In the early years of studying the economics of climate, economists were concerned about the economic costs of moving too quickly [100, 101, 124–126]. As the problem became better understood, a different picture emerged, indicating that the benefits of at least modest climate action should significantly exceed costs from the societal perspective. Subsequent analyses showed that substantially reducing emissions would come at gross societal costs of *at most* a few percent of GDP in coming decades, resulting in the loss of roughly a single year's growth in GDP [127–134]. Further, these analyses showed that delaying action had serious downside risks, as discussed below.

Over time, economic assessments became more sophisticated and started to include important factors that the earliest simple models omitted—factors whose inclusion generally showed that achieving lower emissions would be cheaper, easier, and more beneficial for society than the earlier assessments indicated [135–139]. Models often omitted air quality and other health co-benefits, which are often big enough to fully offset the gross costs of reducing emissions, justifying significant climate action even before considering greenhouse gas externalities [140–145]. Many models omitted bottom-up analysis of market reforms and technology programs, which are important sources of negative net cost (i.e. societally profitable) emissions reduction options [129, 136, 137, 146–151].

Many **benefit–cost models** ignored climate damages for reference cases, assuming economic growth rates in those cases would be unaffected by unrestricted climate change [152]. Many also relied on flawed estimates of climate damages that vastly underestimate the benefits of reducing emissions [153–156]. Standard calculations of discount rates by Nordhaus and others also included a term for business-as-usual economic growth unencumbered by potential climate damages [157], and that assumption led to higher discount rates that make future benefits of climate action appear less valuable than in reality.

Virtually all models ignored the 'long-tail' risks of catastrophic climate change, which lay bare the weaknesses of the benefit–cost framing like no other issue [87, 158]. Benefit–cost models also often ignored the lessons from financial economics on pricing risk and uncertainty [159] and almost always ignored potential cost reductions associated with economies of scale, learning effects, spillovers, network externalities, and irreversibilities [138, 160–167], although sometimes these effects were studied in specific cases [162, 168]. They also almost always ignore people's asymmetrical treatment of losses versus gains (prospect theory), which again biased the results toward delaying mitigation [156].

Some models also omitted benefits from using carbon tax revenues to reduce inefficiencies in the tax system [169], failed to distinguish between endogenous and directed or induced technical change [166, 170, 171], and included only local or regional emissions trading and focused only on carbon dioxide rather than all greenhouse gases [135, 172]. In almost every case, including these factors in the analysis would have shown emissions reductions to be cheaper for society than in the original analysis.

Prematurely excluding cost-competitive mitigation options from analyses like these can raise the apparent cost of reducing emissions. For example, in many places excluding the option of extending the useful life of existing nuclear plants would make achieving emissions goals appear to be more expensive than they would be if this option was kept on the table. How this constraint nets out for a particular technology, policy, or scenario depends on the cost for each option.

Conversely, including options as 'backstop' technologies with optimistic costs and potentials, as has happened in the past, for example, in the case of biomass energy with carbon capture, can make achieving emissions reduction goals appear cheaper than they really would be in practice. On balance, however, the models historically have erred more on the side of making the costs of emissions reductions appear to be more expensive than they really are.

1.6.6 Climate action, equity, and justice

While the costs of climate change fall on all of us, a disproportionate share of those costs will continue fall on the poorest countries, the poorest people, and generations yet to be born [173–176]. The pursuit of equity and justice is central to action on climate, a fact that is uncomfortable for many economists. The economics field has traditionally focused on economic efficiency, not on disparities in income and power relationships, but these issues cannot be ignored in facing the climate challenge.

Climatologist Michael E Mann, writing in his book, *The New Climate War*, wrote:

> …social justice is intrinsic to climate action. Environmental crises, including climate change, disproportionately impact those with the least wealth, the fewest resources, and the least resilience. So simply acting on the climate crisis is acting to alleviate social injustice. It's another compelling reason to institute the systemic changes necessary to avert the further warming of our planet [177].

Some climate policies are more justice-enhancing than others. Unless policies are designed with existing inequities in mind, they risk exacerbating inequality and injustice, so it is not always true that 'acting on the climate crisis is acting to alleviate social injustice' [175, 176, 178]. What is true is that those with the fewest resources will be hit the hardest by climate change [175, 176], climate change is already increasing inequality [176, 179, 180], and society has an obligation to reduce those harms by rapidly reducing emissions and structuring climate policies to address (or at a minimum not exacerbate) existing inequities.

There is some empirical work supporting the differential effects of climate change and other environmental issues on less advantaged groups, for example [176, 181–183], but it's also an intuitively reasonable conclusion. Marginalized people have fewer resources to manage unexpected crises and often live in places susceptible to extreme weather events and environmental pollution. Structural racism exacerbates these inequalities [184, 185].

For example, when extreme temperatures hit, economically disadvantaged communities have less access to well-conditioned private or public spaces. When flooding or hurricanes hit, poorer families have less ability to move to safer ground, either because they are tied to low wage jobs for survival or don't have access to affordable transportation. Communities of color and low-income communities are exposed to more outdoor air pollution than less diverse and more affluent neighborhoods [186], and exposure to air pollution may also be related to increased mortality from COVID-19 and other infections [187]. Many poorer countries are also significantly affected by air and water pollution, and correctly accounting for co-benefits to greenhouse gas emissions reductions should feature prominently in international negotiations on climate targets and commitments [145]. Air pollution from fossil fuels kills millions of people every year [141, 188], and those deaths are borne disproportionately by poorer people.

Climate change is also deeply intertwined with intergenerational justice [69, 189–191]. Those likely to be most affected by a world with a rapidly changing climate are not yet born, while powerful economic interests make their preference for delayed action known loudly in public proceedings and news media around the world.

Another dimension to the question of justice and the climate is the disproportionate effects the wealthy have on emissions [192–194]. The primary beneficiaries of fossil fuel wealth have been rich countries, and rich people everywhere, but they've privatized benefits while socializing costs. That disparity makes it incumbent upon the wealthy to take the lead on climate action. They can afford it, they are historically more to blame for the climate problem than those people who are less well off, and their prominence in society gives them greater ability to influence the opinions of others [195, 196].

As discussed above, humanity's choices for response are threefold: mitigation, adaptation, and suffering. Our decisions on balancing climate mitigation with adaptation and suffering are at their core *moral choices*. How fast and how far we mitigate are not solely questions of economics and are inseparable from questions about equity and justice [173]. The faster and more deeply we reduce emissions and the more we center the correction of existing inequities in our solutions, the less adaptation and suffering we'll do, and the more rapidly justice will be served.

1.6.7 Speed trumps perfection in climate solutions

The seriousness of the climate problem has been obvious to informed observers for at least three decades, but our delay in facing it has virtually guaranteed that little about humanity's response will be optimal. The most important lesson is that we need to get started on rapid emissions reductions as soon as possible. There's no more time to waste.

The IPCC [49], writing in early 2022, emphasized the urgency of responding to the climate problem without delay:

> The cumulative scientific evidence is unequivocal: climate change is a threat to human well-being and planetary health. Any further delay in concerted anticipatory global action on adaptation and mitigation will miss a brief and rapidly closing window of opportunity to secure a livable and sustainable future for all.

The IPCC assigned *Very High Confidence* to this statement, meaning there is little doubt that the statement is true, with multiple, independent, and consistent lines of evidence supporting it [197]. The quotation itself is scientist-speak for 'It's an emergency, so get cracking!' and 'It's warming, it's us, we're sure, it's bad, we can fix it (but we'd better hurry)'.

There are lessons for climate solutions from responding to other kinds of emergencies. Dr Michael Ryan, Executive Director of the World Health Organization, in talking about pandemic response in early 2020, said this:

> Be fast. Have no regrets. You must be the first mover… In emergency response, if you need to be right before you move, you will never win… Speed trumps perfection…The greatest error is not to move. The greatest error is to be paralyzed by the fear of failure[7].

We conclude from this advice:

- Don't obsess about optimality, just move as quickly as you can.
- Don't obsess about feasibility, just move quickly as you can.
- Don't obsess about obstacles, just move quickly as you can.

For climate, just like for pandemic response, speed trumps perfection, and that's the attitude we need to take in addressing this problem now, because it's a real emergency [198, 199].

1.7 Chapter conclusions

We are in a climate emergency, but most people and institutions aren't acting like it. We need to treat climate like the crisis that it is. That means moving more quickly than in normal times, getting started on rapid emissions reductions immediately. There's no more time to waste.

Keeping global surface temperatures from exceeding 1.5 °C above pre-industrial times will not be easy, nor will drawing down GHGs into climate-positive territory. On the contrary, climate change is the biggest collective challenge modern humanity has ever faced.

We'll need to cut absolute global greenhouse emissions in half by 2030, reaching net zero emissions no later than 2040, then remove emissions with climate-positive processes for many decades thereafter. Industrialized nations, including China, should move even more quickly. Aggressive early action in these nations will drive

[7] https://twitter.com/drericding/status/1340997408503853058?s=10

the cost of zero emissions and climate-positive technologies down substantially, benefitting the entire world.

The vast majority of proved fossil fuel reserves will need to be kept in the ground to stabilize the climate. No fossil fuel companies' business plans currently reflect this reality [200], creating what Al Gore called a 'carbon asset bubble' [113]. When this bubble bursts, as it inevitably will, fossil investors will be left holding the bag.

For climate, just like for pandemic response, speed trumps perfection, and that's the attitude we need to take in addressing this problem now. There will always be social and environmental tradeoffs as we scale up climate solutions, but the direct benefits and co-benefits of scaling solutions generally far outweigh the costs, and we already have techniques for minimizing harm while maximizing benefits for nearly everyone. The rest of this book explores tools needed to speed up climate action and truly face the climate challenge.

I'm skeptical that a problem as complex as climate change can be solved by any single branch of science. Technological measures and regulations are important, but equally important is support for education, ecological training and ethics—a consciousness of the commonality of all living beings and an emphasis on shared responsibility.

—Vaclav Havel (*New York Times* 27 September 2007)

Further reading

Burke K D, Williams J W, Chandler M A, Haywood A M, Lunt D J and Otto-Bliesner B L 2018 Pliocene and Eocene provide best analogs for near-future climates *Proc. Natl Acad. Sci.* **115** 13288 https://doi.org/10.1073/pnas.1809600115. This article summarizes current knowledge about past climates going back millions of years.

Dessler A E 2022 *Introduction to Modern Climate Change* (Cambridge: Cambridge University Press). An excellent summary of the latest climate science.

Duane T, Koomey J, Belyeu K and Hausker K 2016 From risk to return: investing in a clean energy economy *Risky Business* 6 December http://riskybusiness.org/fromrisktoreturn/. A business-focused study of reducing emissions in the US energy sector.

Goldstein-Rose S 2020 *The 100% Solution: A Plan for Solving Climate Change* (New York: Melville House). A terrific introduction to the climate problem for non-technical audiences.

Grubler A *et al* 2018 A low energy demand scenario for meeting the 1.5 °C target and sustainable development goals without negative emission technologies *Nat. Energy* **3** 515–27 https://doi.org/10.1038/s41560-018-0172-6. An exemplary aggressive emissions reduction scenario focusing on changes in energy service demands and efficiency.

IPCC 2018 *Global Warming of 1.5°C. An IPCC Special Report on the Impacts of Global Warming of 1.5°C Above Pre-industrial Levels and Related Global Greenhouse Gas Emission Pathways, in the Context of Strengthening the Global Response to the Threat of Climate Change, Sustainable Development, and Efforts to Eradicate Poverty* (Geneva: IPCC) https://www.ipcc.ch/sr15. An interim IPCC report on aggressive emissions scenarios.

IPCC 2021 *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* ed V Masson-Delmotte *et al* (Cambridge: Cambridge University Press) https://www.ipcc.ch/report/sixth-assessment-report-working-group-i/. The latest IPCC report on the science of climate change.

IPCC 2022 *Climate Change 2022: Impacts, Adaptation and Vulnerability–The Working Group II contribution to the Sixth Assessment Report* (Cambridge: Cambridge University Press) https://www.ipcc.ch/report/sixth-assessment-report-working-group-ii/. The latest IPCC report on impacts of and adaptation to climate change.

IPCC 2022 *Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change DUE OUT MARCH 2022* (Cambridge: Cambridge University Press) https://www.ipcc.ch/report/sixth-assessment-report-working-group-3/. The latest IPCC report on mitigation of climate change.

Mann M E and Toles T 2016 *The Madhouse Effect: How Climate Change Denial is Threatening our Planet, Destroying our Politics, and Driving Us Crazy* (New York: Columbia University Press). An entertaining introduction to the climate problem from one of the world's top climate scientists and one of the world's top editorial cartoonists.

Mann M E 2021 *The New Climate War: The Fight to Take Back Our Planet* (New York: PublicAffairs) 12 January. The latest summary of the climate issue from a top climate scientist.

Rockstrom J, Gaffney O, Rogelj J, Meinshausen M, Nakicenovic N and Schellenhuber H J 2017 A roadmap for rapid decarbonization *Science* **355** 1269 https://doi.org/10.1126/science.aah3443. A widely cited article defining the 'carbon law' benchmark of halving emissions every decade.

Sovacool B K, Burke M, Baker L, Kumar Kotikalapudi C and Wlokas H 2017 New frontiers and conceptual frameworks for energy justice *Energy Policy* **105** 677–91 https://doi.org/10.1016/j.enpol.2017.03.005

References

[1] Krause F, Bach W and Koomey J 1989 *From Warming Fate to Warming Limit: Benchmarks to a Global Climate Convention* (El Cerrito, CA: International Project for Sustainable Energy Paths) http://mediafire.com/file/pzwrsyo1j89axzd/Warmingfatetowarminglimitbook.pdf

[2] Krause F, Bach W and Koomey J G 1992 *Energy Policy in the Greenhouse* (New York: Wiley)
[3] Stoddard I *et al* 2021 Three decades of climate mitigation: why haven't we bent the global emissions curve? *Annu. Rev. Environ. Resour.* **46** 653–89
[4] Nicholas K 2021 *Under The Sky We Make: How to be Human in a Warming World* (New York: Putnam)
[5] Haustein K, Allen M R, Forster P M, Otto F E L, Mitchell D M, Matthews H D and Frame D J 2017 A real-time global warming index *Sci. Rep.* **7** 15417
[6] Kaufman D, McKay N, Routson C, Erb M, Dätwyler C, Sommer P S, Heiri O and Davis B 2020 Holocene global mean surface temperature, a multi-method reconstruction approach *Sci. Data* **7** 201
[7] Kaufman D *et al* 2020 A global database of holocene paleotemperature records *Sci. Data* **7** 115
[8] Fogt R L, Sleinkofer A M, Raphael M N and Handcock M S 2022 A regime shift in seasonal total Antarctic sea ice extent in the twentieth century *Nat. Clim. Change* **12** 54–62
[9] Zemp M *et al* 2019 Global glacier mass changes and their contributions to sea-level rise from 1961 to 2016 *Nature* **568** 382–6
[10] Slater T, Hogg A E and Mottram R 2020 Ice-sheet losses track high-end sea-level rise projections *Nat. Clim. Change* **10** 879–81
[11] Slater T, Lawrence I R, Otosaka I N, Shepherd A, Gourmelen N, Jakob L, Tepes P, Gilbert L and Nienow P 2021 Review article: Earth's ice imbalance *Cryosphere* **15** 233–46
[12] Dangendorf S, Hay C, Calafat F M, Marcos M, Piecuch C G, Berk K and Jensen J 2019 Persistent acceleration in global sea-level rise since the 1960s *Nat. Clim. Change* **9** 705–10
[13] Sweet W V *et al* 2022 Global and regional sea level rise scenarios for the United States: updated mean projections and extreme water level probabilities along US coastlines *NOAA Technical Report* NOS 01 National Oceanic and Atmospheric Administration, National Ocean Service, Silver Spring, MD https://oceanservice.noaa.gov/hazards/sealevelrise/sea-levelrise-tech-report.html
[14] Lumpkin R *et al* 2020 Global oceans *Bull. Am. Meteorol. Soc.* **101** S129–84
[15] Kramer R J, He H, Soden B J, Oreopoulos L, Myhre G, Forster P M and Smith C J 2021 Observational evidence of increasing global radiative forcing *Geophys. Res. Lett.* **48** e2020GL091585
[16] McSweeney R 2020 Jet stream: is climate change causing more 'blocking' weather events? *Carbon Brief* 12 June https://carbonbrief.org/jet-stream-is-climate-change-causing-more-blocking-weather-events/
[17] Burke K D, Williams J W, Chandler M A, Haywood A M, Lunt D J and Otto-Bliesner B L 2018 Pliocene and eocene provide best analogs for near-future climates *Proc. Natl Acad. Sci.* **115** 13288
[18] Haq B U 2014 Cretaceous eustasy revisited *Global Planet. Change* **113** 44–58
[19] IPCC 2021 *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* ed V Masson-Delmotte *et al* (Cambridge: Cambridge University Press) https://ipcc.ch/report/sixth-assessment-report-working-group-i/
[20] Friedlingstein P *et al* 2022 Global carbon budget 2021 *Earth Syst. Sci. Data* **14** 1917–2005
[21] Gütschow J, Jeffery M L, Gieseke R, Gebel R, Stevens D, Krapp M and Rocha M 2016 The PRIMAP-hist national historical emissions time series *Earth Syst. Sci. Data* **8** 571–603

[22] Siegenthaler U *et al* 2005 Stable carbon cycle–climate relationship during the late Pleistocene *Science* **310** 1313–7

[23] Petit J R *et al* 1999 Climate and atmospheric history of the past 420,000 years from the Vostok ice core, Antarctica *Nature* **399** 429–36

[24] IPCC 2021 *Summary for Policy Makers. Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* ed V Masson-Delmotte *et al* (Cambridge: Cambridge University Press) https://ipcc.ch/report/sixth-assessment-report-working-group-i/

[25] NAS 2010 *Advancing the Science of Climate Change* (Washington, DC: National Academy of Sciences) http://nap.edu/catalog.php?record_id=12782

[26] Cook J *et al* 2016 Consensus on consensus: a synthesis of consensus estimates on human-caused global warming *Environ. Res. Lett.* **11** 048002

[27] IPCC 2007 *Climate Change 2007: The Physical Science Basis – Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change* ed S Solomon *et al* (Cambridge: Cambridge University Press) http://ipcc.ch/publications_and_data/publications_and_data_reports.shtml

[28] IPCC 2013 *Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change* ed T F Stocker *et al* (Cambridge: Cambridge University Press) http://climatechange2013.org

[29] WMO 2021 Climate change indicators and impacts worsened in 2020 *State of the Global Climate 2020* WMO-No. 1264 World Meteorological Association, Geneva https://public.wmo.int/en/media/press-release/climate-change-indicators-and-impacts-worsened-2020

[30] Basu S, Lehman S J, Miller J B, Andrews A E, Sweeney C, Gurney K R, Xu X, Southon J and Tans P P 2020 Estimating US fossil fuel CO_2 emissions from measurements of 14 °C in atmospheric CO_2 *Proc. Natl Acad. Sci.* **117** 13300

[31] Broecker W S 1975 Climatic change: are we on the brink of a pronounced global warming? *Science* **189** 460–3

[32] Exxon 1982 CO_2 Greenhouse Effect *Technical review* 82EAP 256 Exxon Corporation, Florham Park, NJ https://insideclimatenews.org/wp-content/uploads/2015/09/1982-Exxon-Primer-on-CO2-Greenhouse-Effect.pdf

[33] Jackson R 2020 Eunice Foote, John Tyndall and a question of priority *R. Soc. J. History Sci.* **74** 105–18

[34] Weart S R 2008 *The Discovery of Global Warming* (Cambridge, MA: Harvard University Press)

[35] Rahmstorf S, Cazenave A, Church J A, Hansen J E, Keeling R F, Parker D E and Somerville R C J 2007 Recent climate observations compared to projections *Science* **316** 709

[36] Harries J E, Brindley H E, Sagoo P J and Bantges R J 2001 Increases in greenhouse forcing inferred from the outgoing longwave radiation spectra of the Earth in 1970 and 1997 *Nature* **410** 355–7

[37] Philipona R, Durr B, Marty C, Ohmura A and Wild M 2004 Radiative forcing—measured at Earth's surface—corroborate the increasing greenhouse effect *Geophys. Res. Lett.* **31** L03202

[38] Murphy D M, Solomon S, Portmann R W, Rosenlof K H, Forster P M and Wong T 2009 An observationally based energy balance for the Earth since 1950 *J. Geophys. Res.* **114** D17107

[39] Domingues C M, Church J A, White N J, Gleckler P J, Wijffels S E, Barker P M and Dunn J R 2008 Improved estimates of upper-ocean warming and multi-decadal sea-level rise *Nature* **453** 1090–3
[40] Strong A, Levin K and Tirpak D 2011 Climate science: major new discoveries *Issue Brief* World Resources Institute, Washington, DC http://wri.org/publication/climate-science
[41] Xu Y, Ramanathan V and Victor D G 2018 Global warming will happen faster than we think *Nature* **564** 30–2
[42] Doerr S H and Santín C 2016 Global trends in wildfire and its impacts: perceptions versus realities in a changing world *Phil. Trans. R. Soc.* B **371** 20150345
[43] Burke M, Driscoll A, Heft-Neal S, Xue J, Burney J and Wara M 2021 The changing risk and burden of wildfire in the United States *Proc. Natl Acad. Sci.* **118** e2011048118
[44] Park T, Ganguly S, Tømmervik H, Euskirchen E S, Høgda K-A, Karlsen S R, Brovkin V, Nemani R R and Myneni R B 2016 Changes in growing season duration and productivity of northern vegetation inferred from long-term remote sensing data *Environ. Res. Lett.* **11** 084001
[45] Ortiz-Bobea A, Ault T R, Carrillo C M, Chambers R G and Lobell D B 2021 Anthropogenic climate change has slowed global agricultural productivity growth *Nat. Clim. Change* **11** 306–12
[46] Henson R 2011 *The Rough Guide to Climate Change* 3rd edn (London: Rough Guides)
[47] Xu C, Kohler T A, Lenton T M, Svenning J-C and Scheffer M 2020 Future of the human climate niche *Proc. Natl Acad. Sci.* **117** 11350–5
[48] Zeppetello L R V, Raftery A E and Battisti D S 2022 Probabilistic projections of increased heat stress driven by climate change *Commun. Earth Environ.* **3** 183
[49] IPCC 2022 *Climate Change 2022: Impacts, Adaptation and Vulnerability—The Working Group II Contribution to the Sixth Assessment Report* (Cambridge: Cambridge University Press) https://ipcc.ch/report/sixth-assessment-report-working-group-ii/
[50] Fricko O *et al* 2017 The marker quantification of the shared socioeconomic pathway 2: a middle-of-the-road scenario for the 21st century *Global Environ. Change* **42** 251–67
[51] Hausfather Z and Peters G P 2020 Emissions—the 'business as usual' story is misleading *Nature* **577** 618–20
[52] Hausfather Z and Forster P 2021 Analysis: Do Cop26 Promises Keep Global Warming Below 2 °C? *Carbon Brief* 10 November https://carbonbrief.org/analysis-do-cop26-promises-keep-global-warming-below-2c
[53] Sognnaes I *et al* 2021 A multi-model analysis of long-term emissions and warming implications of current mitigation effortsNat. Clim. Change **11** 1055–62
[54] Moore F C, Lacasse K, Mach K J, Shin Y A, Gross L J and Beckage B 2022 Determinants of emissions pathways in the coupled climate–social system *Nature* **603** 103–11
[55] Koomey J G 2012 *Cold Cash, Cool Climate: Science-Based Advice for Ecological Entrepreneurs* (El Dorado Hills, CA: Analytics)
[56] Ritchie J and Dowlatabadi H 2017 The 1000 GtC coal question: are cases of vastly expanded future coal combustion still plausible? *Energy Econ.* **65** 16–31
[57] Quiggin D, Meyer K D, Hubble-Rose L and Froggatt A 2021 Climate Change Risk Assessment 2021: The Risks Are Compounding, and Without Immediate Action the Impacts Will Be Devastating *Environment and Society Programme Research Paper* Chatham House, London https://chathamhouse.org/sites/default/files/2021-09/2021-09-14-climate-change-risk-assessment-quiggin-et-al.pdf

[58] IPCC 2021 Weather and climate extreme events in a changing climate *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* ed V Masson-Delmotte *et al* (Cambridge: Cambridge University Press) ch 11 https://ipcc.ch/report/sixth-assessment-report-working-group-i/
[59] Grübler A *et al* 2018 A low energy demand scenario for meeting the 1.5 °C target and sustainable development goals without negative emission technologies *Nat. Energy* **3** 515–27
[60] US Climate Change Science Program 2008 *Weather and Climate Extremes in a Changing Climate—Regions of Focus: North America, Hawaii, Caribbean, and US Pacific Islands* (Washington, DC: US Climate Change Science Program and the Subcommittee on Global Change Research) https://downloads.globalchange.gov/sap/sap3-3/sap3-3-final-all.pdf
[61] Dai A 2011 Drought under global warming: a review *WIREs Climate Change* **2** 45–65
[62] Rahmstorf S and Coumou D 2011 Increase of extreme events in a warming world *Proc. Natl Acad. Sci.* **108** 17905–9
[63] Romm J 2011 The next dust bowl *Nature* **478** 450–1
[64] Singh J, Ashfaq M, Skinner C B, Anderson W B, Mishra V and Singh D 2022 Enhanced risk of concurrent regional droughts with increased ENSO variability and warming *Nat. Clim. Change* **12** 163–70
[65] Hansen J, Sato M and Ruedy R 2012 *The New Climate Dice: Public Perception of Climate Change* (New York: National Aeronautics and Space Administration, Goddard Institute for Space Studies) https://www.giss.nasa.gov/research/briefs/2012_hansen_17/
[66] Ceballos G, Ehrlich P R and Dirzo R 2017 Biological annihilation via the ongoing sixth mass extinction signaled by vertebrate population losses and declines *Proc. Natl Acad. Sci.* **114** E6089
[67] IPBES 2019 *Summary for Policymakers of the Global Assessment Report on Biodiversity and Ecosystem Services, IPBES Plenary at its Seventh Session* (Paris: Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services)
[68] Balint M, Domisch S, Engelhardt C H M, Haase P, Lehrian S, Sauer J, Theissinger K, Pauls S U and Nowak C 2011 Cryptic biodiversity loss linked to global climate change *Nature Clim. Change* **1** 313–8
[69] Thompson A 2022 How climate change will hit younger generations *Sci. Am.* 1 February p 76
[70] Anderegg W R L, Abatzoglou J T, Anderegg L D L, Bielory L, Kinney P L and Ziska L 2021 Anthropogenic climate change is worsening North American pollen seasons *Proc. Natl Acad. Sci.* **118** e2013284118
[71] Carlson C J, Albery G F, Merow C, Trisos C H, Zipfel C M, Eskew E A, Olival K J, Ross N and Bansal S 2022 Climate change increases cross-species viral transmission risk *Nature* **607** 555–62
[72] Yong E 2022 We created the 'Pandemicene': by completely rewiring the network of animal viruses, climate change is creating a new age of infectious dangers *The Atlantic* 28 April https://theatlantic.com/science/archive/2022/04/how-climate-change-impacts-pandemics/629699/
[73] Grinsted A and Christensen J H 2021 The transient sensitivity of sea level rise *Ocean Sci.* **17** 181–6
[74] Rahmstorf S 2007 A semi-empirical approach to projecting future sea-level rise *Science* **315** 368–70

[75] Vermeer M and Rahmstorf S 2009 Global sea level linked to global temperature *Proc. Natl Acad. Sci.* **106** 21527–32

[76] Ridgwell A and Schmidt D N 2010 Past constraints on the vulnerability of marine calcifiers to massive carbon dioxide release *Nat. Geosci.* **3** 196–200

[77] Sale P 2011 *Our Dying Planet: An Ecologist's View of the Crisis We Face* (Berkeley, CA: University of California Press)

[78] Dias B B, Hart M B, Smart C W and Hall-Spencer J M 2010 Modern seawater acidification: the response of foraminifera to high-CO_2 conditions in the Mediterranean Sea *J. Geol. Soc.* **167** 843–6

[79] Rantanen M, Karpechko A Y, Lipponen A, Nordling K, Hyvärinen O, Ruosteenoja K, Vihma T and Laaksonen A 2022 The Arctic has warmed nearly four times faster than the globe since 1979 *Commun. Earth Environ.* **3** 168

[80] Lenton T M, Rockström J, Gaffney O, Rahmstorf S, Richardson K, Steffen W and Schellnhuber H J 2019 Climate tipping points—too risky to bet against *Nature* **575** 592–5

[81] Kemp L *et al* 2022 Climate endgame: exploring catastrophic climate change scenarios *Proc. Natl Acad. Sci.* **119** e2108146119

[82] Simpson R D 2022 How do we price an unknowable risk? *Issues in Science and Technology* **38** (Winter) 47–50 https://issues.org/social-cost-of-carbon-economics-simpson/

[83] Orihuela-Pinto B, England M H and Taschetto A S 2022 Interbasin and interhemispheric impacts of a collapsed Atlantic overturning circulationNat. Clim. Change **12** 558–65

[84] Caesar L, McCarthy G D, Thornalley D J R, Cahill N and Rahmstorf S 2021 Current Atlantic meridional overturning circulation weakest in last millennium *Nat. Geosci.* **14** 118–20

[85] Rahmstorf S, Box J E, Feulner G, Mann M E, Robinson A, Rutherford S and Schaffernicht E J 2015 Exceptional twentieth-century slowdown in Atlantic Ocean overturning circulation *Nat. Clim. Change* **5** 475–80

[86] Jackson L C, Biastoch A, Buckley M W, Desbruyères D G, Frajka-Williams E, Moat B and Robson J 2022 The evolution of the North Atlantic meridional overturning circulation since 1980 *Nat. Rev. Earth Environ.* **3** 241–54

[87] Weitzman M L 2009 On modeling and interpreting the economics of catastrophic climate change *Rev. Econ. Stat.* **91** 1–19

[88] Weitzman M L 2010 What is the 'damages function' for global warming—and what difference might it make? *Clim. Change Econ.* **1** 57–69

[89] Weitzman M L 2011 Fat-tailed uncertainty in the economics of catastrophic climate change *Rev. Environ. Econ. Policy* **5** 275–92

[90] Wagner G and Weitzman M W 2016 *Climate Shock: The Economic Consequences of a Hotter Planet* (Princeton, NJ: Princeton University Press)

[91] IPCC 2022 *Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge: Cambridge University Press) https://ipcc.ch/report/sixth-assessment-report-working-group-3/

[92] Allen M R, Friedlingstein P, Girardin C A J, Jenkins S, Malhi Y, Mitchell-Larson E, Peters G P and Rajamani L 2022 Net zero: science, origins, and implications *Ann. Rev. Environ. Res.* **47** 849–87

[93] Matthews H D, Gillett N P, Stott P A and Zickfeld K 2009 The proportionality of global warming to cumulative carbon emissions *Nature* **459** 829–32

[94] Zickfeld K, Eby M, Matthews H D and Weaver A J 2009 Setting cumulative emissions targets to reduce the risk of dangerous climate change *Proc. Natl Acad. Sci.* **106** 16129
[95] Ricke K L and Caldeira K 2014 Maximum warming occurs about one decade after a carbon dioxide emission *Environ. Res. Lett.* **9** 124002
[96] Matthews H D and Caldeira K 2008 Stabilizing climate requires near-zero emissions *Geophys. Res. Lett.* **35** L04705
[97] Lahn B 2020 A history of the global carbon budget *WIREs Clim. Change.* **11** e636
[98] Koomey J 2013 Moving beyond benefit-cost analysis of climate change *Environ. Res. Lett.* **8** 041005
[99] Baumol W J 1972 On taxation and the control of externalities *Am. Econ. Rev.* **62** 307–22 http://jstor.org/stable/1803378
[100] Nordhaus W D 1977 Economic growth and climate: the carbon dioxide problem *Am. Econ. Rev.* **67** 341–6 https://www.jstor.org/stable/1815926
[101] Nordhaus W D 1979 *The Efficient Use of Energy Resources* (New Haven, CT: Yale University Press)
[102] Caldeira K, Jain A K and Hoffert M I 2003 Climate sensitivity uncertainty and the need for energy without CO_2 emission *Science* **299** 2052–4
[103] Meinshausen M, Meinshausen N, Hare W, Raper S C B, Frieler K, Knutti R, Frame D J and Allen M R 2009 Greenhouse-gas emission targets for limiting global warming to 2 °C *Nature* **458** 1158–62
[104] IEA 2010 *World Energy Outlook 2010* (Paris: International Energy Agency, Organization for Economic Cooperation and Development) http://worldenergyoutlook.org/
[105] IEA 2011 *World Energy Outlook 2011* (Paris: International Energy Agency, Organization for Economic Cooperation and Development) http://worldenergyoutlook.org/
[106] IEA 2012 *World Energy Outlook 2012* (Paris: International Energy Agency, Organization for Economic Cooperation and Development) http://worldenergyoutlook.org/
[107] IEA 2020 *World Energy Outlook 2020* (Paris: International Energy Agency, Organization for Economic Cooperation and Development) http://worldenergyoutlook.org/
[108] Luderer G, Pietzcker R C, Bertram C, Kriegler E, Meinshausen M and Edenhofer O 2013 Economic mitigation challenges: how further delay closes the door for achieving climate targets *Environ. Res. Lett.* **8** 034033
[109] Rockström J, Gaffney O, Rogelj J, Meinshausen M, Nakicenovic N and Schellnhuber H J 2017 A roadmap for rapid decarbonization *Science* **355** 1269
[110] IPCC 2018 *Global Warming of 1.5 °C. An IPCC Special Report on the Impacts of Global Warming of 1.5 °C above Pre-industrial Levels and Related Global Greenhouse Gas Emission Pathways, in the Context of Strengthening the Global Response to the Threat of Climate Change, Sustainable Development, and Efforts to Eradicate Poverty* (Geneva: IPCC) https://ipcc.ch/sr15
[111] Rogelj J, Meinshausen M, Sedláček J and Knutti R 2014 Implications of potentially lower climate sensitivity on climate projections and policy *Environ. Res. Lett.* **9** 031003
[112] McKibben B 2012 Global warming's terrifying new math *Rolling Stone Mag.* 19 July https://www.rollingstone.com/politics/politics-news/global-warmings-terrifying-new-math-188550/
[113] Gore A and Blood D 2013 The coming carbon asset bubble *The Wall Street Journal* 29 October http://online.wsj.com/news/articles/SB100014240527023046551045791636634643 39836?mod=hp_opinion

[114] Matthews H D *et al* 2020 Opportunities and challenges in using remaining carbon budgets to guide climate policy *Nat. Geosci.* **13** 769–79
[115] Rogelj J, Meinshausen M, Schaeffer M, Knutti R and Riahi K 2015 Impact of short-lived non-CO_2 mitigation on carbon budgets for stabilizing global warming *Environ. Res. Lett.* **10** 075001
[116] Rogelj J, Reisinger A, McCollum D L, Knutti R, Riahi K and Meinshausen M 2015 Mitigation choices impact carbon budget size compatible with low temperature goals *Environ. Res. Lett.* **10** 075003
[117] Tokarska K and Matthews D 2021 Refining the remaining 1.5 °C 'carbon budget' *Carbon Brief* 19 January https://carbonbrief.org/guest-post-refining-the-remaining-1-5c-carbon-budget
[118] Matthews H D, Tokarska K B, Rogelj J, Smith C J, MacDougall A H, Haustein K, Mengis N, Sippel S, Forster P M and Knutti R 2021 An integrated approach to quantifying uncertainties in the remaining carbon budget *Commun. Earth Environ.* **2** 7
[119] MacDougall A H, Zickfeld K, Knutti R and Matthews H D 2015 Sensitivity of carbon budgets to permafrost carbon feedbacks and non-CO_2 forcings *Environ. Res. Lett.* **10** 125003
[120] Herman A 2012 *Freedom's Forge: How American Business Produced Victory in World War II* (New York: Random House)
[121] Lovins A B *et al* 2011 *Reinventing Fire: Bold Business Solutions for the New Energy Era* (White River Junction, VT: Chelsea Green) https://rmi.org/insights/reinventing-fire/
[122] Koomey J G, Matthews H S and Williams E 2013 Smart everything: will intelligent systems reduce resource use? *Ann. Rev. Environ. Res.* **38** 311–43
[123] BGR 2020 *BGR Energy Study 2019—Data and Developments Concerning German and Global Energy Supplies* (Hannover: Federal Institute for Geosciences and Natural Resources) https://www.bgr.bund.de/EN/Themen/Energie/Produkte/energy_study_2019_-summary_en.html
[124] Nordhaus W D 1992 An optimal transition path for controlling greenhouse gases *Science* **258** 1315
[125] Wigley T M L, Richels R and Edmonds J A 1996 Economic and environmental choices in the stabilization of atmospheric CO_2 concentrations *Nature* **379** 240–43
[126] Manne A and Richels R 1997 On stabilizing CO_2 concentrations—cost-effective emission reduction strategies *Environ. Model. Assessment* **2** 251–65
[127] IPCC 2014 *Climate Change 2014: Mitigation of Climate Change. Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change* ed O Edenhofer *et al* (Cambridge: Cambridge University Press) https://www.ipcc.ch/report/ar5/wg3/
[128] Kriegler E *et al* 2014 The role of technology for achieving climate policy objectives: overview of the EMF 27 study on global technology and climate policy strategies *Clim. Change* **123** 353–67
[129] Krause F, Haites E, Howarth R and Koomey J G 1993 *Cutting Carbon Emissions–Burden or Benefit? The Economics of Energy-Tax and Non-Price Policies* (El Cerrito, CA: International Project for Sustainable Energy Paths) http://mediafire.com/file/y4r0n4vcvv4257s/cuttingCemissionsburdenorbenefitbook.pdf
[130] Duane T, Koomey J, Belyeu K and Hausker K 2016 From risk to return: investing in a clean energy economy *Risky Business* http://riskybusiness.org/fromrisktoreturn/

[131] Williams J H, Jones R A, Haley B, Kwok G, Hargreaves J, Farbes J and Torn M S 2021 Carbon-neutral pathways for the United States *AGU Adv.* **2** e2020AV000284

[132] USGCRP 2018 *Impacts, Risks, and Adaptation in the United States: Fourth National Climate Assessment* vol 2 (Washington, DC: US Global Change Research Program) https://nca2018.globalchange.gov

[133] DeCanio S J 2003 *Economic Models of Climate Change: A Critique* (Basingstoke: Palgrave-Macmillan)

[134] Lovins A B, Lovins H, Krause F and Bach W 1981 *Least-Cost Energy: Solving the CO_2 Problem* (Andover, MA: Brick House)

[135] Laitner J A, Decanio S J, Koomey J G and Sanstad A H 2003 Room for improvement: increasing the value of energy modeling for policy analysis *Utilities Policy* **11** 87–94

[136] Krause F, Koomey J and Olivier D 2000 *Cutting Carbon Emissions While Making Money: Climate Saving Energy Strategies for the European Union (Executive Summary for Volume II, Part 2 of Energy Policy in the Greenhouse)* (El Cerrito, CA: International Project for Sustainable Energy Paths) http://mediafire.com/file/bxdgkkb2d5rcjh1/ipsepkyotocosts_eu-report.pdf

[137] Krause F, Baer P and DeCanio S 2001 *Cutting Carbon Emissions at a Profit: Opportunities for the US* (El Cerrito, CA: International Project for Sustainable Energy Paths) http://mediafire.com/file/0aro7bj2d7kqk8w/ipsepcutcarbon_us.pdf

[138] IPCC 2007 *Climate Change 2007: Mitigation of Climate Change—Contribution of Working Group III to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change* ed B O Metz *et al* (Cambridge: Cambridge University Press) http://ipcc.ch/publications_and_data/publications_and_data_reports.shtml

[139] Williams J H, DeBenedictis A, Ghanadan R, Mahone A, Moore J, Morrow W R, Price S and Torn M S 2012 The technology path to deep greenhouse gas emissions cuts by 2050: the pivotal role of electricity *Science* **335** 53–9

[140] Nemet G F, Holloway T and Meier P 2010 Implications of incorporating air-quality co-benefits into climate change policymaking *Environ. Res. Lett.* **5** 014007

[141] Vohra K, Vodonos A, Schwartz J, Marais E A, Sulprizio M P and Mickley L J 2021 Global mortality from outdoor fine particle pollution generated by fossil fuel combustion: results from GEOS-Chem *Environ. Res.* **195** 110754

[142] Shindell D 2020 *Health and Economic Benefits of a 2 °C Climate Policy: Testimony by Professor Drew Shindell* (Washington, DC: US House of Representatives) https://oversight.house.gov/sites/democrats.oversight.house.gov/files/Testimony%20Shindell.pdf

[143] Roberts D 2020 Air pollution is much worse than we thought: ditching fossil fuels would pay for itself through clean air alone *Vox* 12 August https://vox.com/energy-and-environment/2020/8/12/21361498/climate-change-air-pollution-us-india-china-deaths

[144] Cohen A J *et al* 2017 Estimates and 25-year trends of the global burden of disease attributable to ambient air pollution: an analysis of data from the global burden of diseases study 2015 *Lancet* **389** 1907–18

[145] Scovronick N, Anthoff D, Dennig F, Errickson F, Ferranna M, Peng W, Spears D, Wagner F and Budolfson M 2021 The importance of health co-benefits under different climate policy cooperation frameworks *Environ. Res. Lett.* **16** 055027

[146] Brown M A, Levine M D, Short W and Koomey J G 2001 Scenarios for a clean energy future *Energy Policy* **29** 1179–96

[147] Gumerman E, Koomey J G and Brown M A 2001 Strategies for cost-effective carbon reductions: a sensitivity analysis of alternative scenarios *Energy Policy* **29** 1313–23

[148] Sanstad A H, DeCanio S, Boyd G and Koomey J G 2001 Estimating bounds on the economy-wide effects of the CEF policy scenarios *Energy Policy* **29** 1299–312

[149] Lovins A B, Ürge-Vorsatz D, Mundaca L, Kammen D M and Glassman J W 2019 Recalibrating climate prospects *Environ. Res. Lett.* **14** 120201

[150] DeCanio S 1993 Barriers within firms to energy-efficient investments *Energy Policy* **21** 906–14

[151] DeCanio S J 1998 The efficiency paradox: bureaucratic and organizational barriers to profitable energy-saving investments *Energy Policy* **26** 441–54

[152] Bastien-Olvera B A 2019 Business-as-usual redefined: energy systems under climate-damaged economies warrant review of nationally determined contributions *Energy* **170** 862–8

[153] Moore F C and Delavane B D 2015 Temperature impacts on economic growth warrant stringent mitigation policy *Nat. Clim. Change* **5** 127–31

[154] Moore F C, Baldos U, Hertel T and Diaz D 2017 New science of climate change impacts on agriculture implies higher social cost of carbon *Nat. Commun.* **8** 1607

[155] Ackerman F and Stanton E A 2012 Climate risks and carbon prices: revising the social cost of carbon *Economics* **6** 1–25

[156] Kulkarni S, Hof A, van der Wijst K-I and van Vuuren D 2022 Disutility of climate change damages warrants much stricter climate targets *Research Square* https://doi.org/10.21203/rs.3.rs-1788130/v1

[157] Mann G 2022 Check your spillover *London Rev. Books* **44**(10 February) https://lrb.co.uk/the-paper/v44/n03/geoff-mann/check-your-spillover

[158] Cai Y and Lontzek T S 2018 The social cost of carbon with economic and climate risks *J. Political Econ.* **127** 2684–734

[159] Daniel K D, Litterman R B and Wagner G 2019 Declining CO_2 price paths *Proc. Natl Acad. Sci.* **116** 20886

[160] Arthur W B 1990 Positive feedbacks in the economy *Sci. Am.* February pp 92–9

[161] Arthur W B 1994 *Increasing Returns and Path Dependence in the Economy* (Ann Arbor, MI: University of Michigan Press)

[162] Gritsevskyi A and Nakicenovic N 2000 Modeling uncertainty of induced technological change *Energy Policy* **28** 907–21

[163] Rao S, Keppo I and Riahi K 2006 Importance of technological change and spillovers in long-term climate policy *Energy J.* **27** 25–42

[164] Arrow K J 1962 The economic implications of learning by doing *Rev. Econ. Studies* **29** 155–73

[165] Vogt-Schilb A, Meunier G and Hallegatte S 2018 When starting with the most expensive option makes sense: optimal timing, cost and sectoral allocation of abatement investment *J. Environ. Econ. Manag.* **88** 210–33

[166] Fischer C and Newell R G 2008 Environmental and technology policies for climate mitigation *J. Environ. Econ. Manag.* **55** 142–62

[167] Sharpe S and Lenton T M 2021 Upward-scaling tipping cascades to meet climate goals: plausible grounds for hope *Climate Policy* **21** 421–33

[168] Nordhaus W D 2014 The perils of the learning model for modeling endogenous technological change *Energy J.* **35** 1–13

[169] Hamond M J, DeCanio S J, Duxbury P, Sanstad A H and Stinson C H 1997 Tax waste, not work *Challenge* **40** 53–62
[170] Acemoglu D, Aghion P, Bursztyn L and Hemous D 2012 The environment and directed technical change *Am. Econ. Rev.* **102** 131–66
[171] Acemoglu D, Akcigit U, Hanley D and Kerr W 2016 Transition to clean technology *J. Political Econ.* **124** 52–104
[172] de la Chesnaye F C and Weyant J P 2006 EMF 21 multi-greenhouse gas mitigation and climate policy *Energy J. Spec. Issue* **27**
[173] Mohai P, Pellow D and Roberts J T 2009 Environmental justice *Ann. Rev. Environ. Res.* **34** 405–30
[174] Sovacool B K, Burke M, Baker L, Kotikalapudi C K and Wlokas H 2017 New frontiers and conceptual frameworks for energy justice *Energy Policy* **105** 677–91
[175] OECD 2021 The inequalities–environment nexus *OECD Green Growth Papers* No. 2021/01 (Paris: OECD Publishing)
[176] Islam S N and Winkel J 2017 Climate Change and Social Inequality *DESA Working Paper* No. 152 ST/ESA/2017/DWP/152 United Nations, Department of Economic and Social Affairs https://un.org/esa/desa/papers/2017/wp152_2017.pdf
[177] Mann M E 2021 *The New Climate War: The Fight to Take Back Our Planet* (New York: Public Affairs)
[178] Shi L and Moser S 2021 Transformative climate adaptation in the United States: trends and prospects *Science* **372** eabc805
[179] Diffenbaugh N S and Burke M 2019 Global warming has increased global economic inequality *Proc. Natl Acad. Sci.* **116** 9808–13
[180] Zheng Y, Steven J D, Persad G G and Caldeira K 2020 Climate effects of aerosols reduce economic inequality *Nat. Clim. Change* **10** 220–4
[181] Hsiang S *et al* 2017 Estimating economic damage from climate change in the United States *Science* **356** 1362
[182] Tessum C W, Paolella D A, Chambliss S E, Apte J S, Hill J D and Marshall J D 2021 $PM_{2.5}$ polluters disproportionately and systemically affect people of color in the United States *Sci. Adv.* **7** eabf4491
[183] UNEP 2021 *Neglected: Environmental Justice Impacts of Marine Litter and Plastic Pollution* (Nairobi: United Nations Environment Program) https://unep.org/resources/report/neglected-environmental-justice-impacts-marine-litter-and-plastic-pollution
[184] Bullard R D 2008 *Dumping in Dixie: Race, Class, and Environmental Quality* 3rd edn (Boulder, CO: Westview)
[185] Bullard R D (ed) 1999 *Confronting Environmental Racism: Voices from the Grassroots* (Boston, MA: South End)
[186] Zou B, Peng F, Wan N, Mamady K and Wilson G J 2014 Spatial cluster detection of air pollution exposure inequities across the United States *PLoS One* **9** e91917
[187] Petroni M, Hill D, Younes L, Barkman L, Howard S, Howell I B, Mirowsky J and Collins M B 2020 Hazardous air pollutant exposure as a contributing factor to COVID-19 mortality in the United States *Environ. Res. Lett.* **15** 0940a9
[188] Errigo I M *et al* 2020 Human health and economic costs of air pollution in Utah: an expert assessment *Atmosphere* **11** 1238

[189] Howarth R B 2011 Intergenerational justice *The Oxford Handbook of Climate Change and Society* ed J S Dryzek, R B Norgaard and D Schlosberg (Oxford: Oxford University Press) pp 338–52 ch 23
[190] Robinson K S 2009 Time to end the multigenerational Ponzi scheme *What Matters* 22 February 22 https://web.archive.org/web/20120705215631/http://whatmatters.mckinseydigital.com/climate_change/time-to-end-the-multigenerational-ponzi-scheme
[191] Rezai A, Duncan K F and Taylor L 2012 Global warming and economic externalities *Econ. Theory* **49** 329–51
[192] Gore T 2020 *Confronting Carbon Inequality: Putting Climate Justice at the Heart of the COVID-19 Recovery* (Nairobi: Oxfam) https://oxfam.org/en/research/confronting-carbon-inequality
[193] Gore T 2015 *Extreme Carbon Inequality: Why the Paris Climate Deal Must Put the Poorest, Lowest Emitting and Most Vulnerable People First* (Nairobi: Oxfam) https://policy-practice.oxfam.org/resources/extreme-carbon-inequality-why-the-paris-climate-deal-must-put-the-poorest-lowes-582545/
[194] Wiedmann T, Lenzen M, Keyßer L T and Steinberger J K 2020 Scientists' warning on affluence *Nat. Commun.* **11** 3107
[195] Nielsen K S, Nicholas K A, Creutzig F, Dietz T and Stern P C 2021 The role of high-socioeconomic-status people in locking in or rapidly reducing energy-driven greenhouse gas emissions *Nat. Energy* **6** 1011–6
[196] Otto I M, Kim K M, Dubrovsky N and Lucht W 2019 Shift the focus from the super-poor to the super-rich *Nat. Clim. Change* **9** 82–4
[197] Mastrandrea M D *et al* 2010 *Guidance Note for Lead Authors of the IPCC Fifth Assessment Report on Consistent Treatment of Uncertainties, IPCC Cross-Working Group Meeting on Consistent Treatment of Uncertainties Jasper Ridge, CA, USA* (Geneva: Intergovernmental Panel on Climate Change) https://ipcc.ch/site/assets/uploads/2017/08/AR5_Uncertainty_Guidance_Note.pdf
[198] Fischetti M 2021 We are living in a climate emergency, and we're going to say so *Sci. Am.* 12 April https://scientificamerican.com/article/we-are-living-in-a-climate-emergency-and-were-going-to-say-so/
[199] Hickel J 2021 What would it look like if we treated climate change as an actual emergency? *Current Affairs* 15 November https://currentaffairs.org/2021/11/what-would-it-look-like-if-we-treated-climate-change-as-an-actual-emergency
[200] Kühne K, Bartsch N, Tate R D, Higson J and Habet A 2022 'Carbon bombs'—mapping key fossil fuel projects *Energy Policy* **166** 112950